SOP Number and version: SOP-TL-NAAB.v01

SOP Title: **Nuclease Activity Assay for Bacteria**

Table 1. SOP current version review and approval table

	NAME	TITLE	SIGNATURE	DATE
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Approval	Luiza I. Hernandez	COO		09/01/2025

Effective Date:	**09/01/2025**
Review Date	09/01/2025

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1. Overview and background

Nuclease Activity Assay (NAA) is a method to investigate the activity of nuclease enzyme(s) in a sample (e.g., bacteria culture, tissue samples, recombinant enzymes). Theralia is the pioneer of using nuclease activity as biomarker for developing targeted therapeutics. Therefore, accurate determination and characterization of nuclease activity is usually initial and a crucial step for our activities.

2. Scope

In this SOP, Nuclease Activity Assay used for bacterial nucleases in Theralia R&D Laboratories located at Linköping University (Campus Valla, B-huset, Rum 2E:686) will be explained.

This SOP covers the sections below.

- Sample preparation
- Performing NAA
- Evaluation of results

3. Procedure

3.1. Sample preparation

- Always use a fresh agar culture on a non-selective media (e.g., Tryptone Soya Agar) to start.

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- Take a single colony and inoculate into a flask with 25-50 mL broth media (e.g., Tryptone Soya Broth) according to the requirements of bacteria used. For initial screening, different broth media may need to be tested.
- Incubate at 36±1°C for 24 hours. Testing different time points (e.g., 12 hours culture) maybe needed for different bacteria.
- Some nucleases (e.g., micrococcal nuclease) may require an additional subculture for obtaining a better nuclease activity. Explore if this is the case by diluting the liquid culture 1/500 using same broth media and incubating again at 36±1°C for 24 hours. Testing different time points (e.g., 12 hours culture) maybe needed for different bacteria.
- Distribute the liquid culture unto centrifuge tubes.
- Centrifuge at 6000 rpm for 20 minutes.
- Separate the supernatant and store at -20°C until use to test **secreted nuclease activity**.
- Wash the pellet (resuspend and centrifuge) two times using PBS.
- Resuspend the pellet in 100 µL PBS to test **cell-wall associated nuclease activity**.
- Keep this cell suspension at 4°C until use up to 24 hours.
- Use an aliquot of cell suspension to disrupt bacterial cells using a chemical (lysozyme) or physical method (homogenization) according to the manufacturer`s recommendations if **intracellular nuclease activity** will also be tested in addition to cell-wall associated nuclease activity.
- Store this preparation at -20°C until use.

3.2. Probe preparation

Fluorescent oligonucleotide probes are used to determine nuclease activity in Theralia R&D Laboratories. The probes are received in lyophilized form. Oligonucleotide concentration of the stock solutions should be 500 pmol/µL.

- Read the datasheet of the probes received carefully.
- Add Tris-EDTA (µL) in the double amount of the oligonucleotide yield (nmol)
  For example, if the yield is 45 nmol, add 90 µL Tris-EDTA.
- This will make the stock solution of 500 pmol/µL.
- Prepare 2 µL aliquots of the stock solution in Eppendorf tubes and store at -80°C together with the screwed capped stock vial.
- Prepare a working solution by diluting an aliquot of the stock solution 1/10 using PBS (different buffers such as sterilized ultrapure water can also be tested) making the concentration 50 pmol/µL.
- Store at -20°C.

3.2. Performing NAA

NAA can be performed in a wide range of experimental conditions related to different requirements of nucleases for the catalytic activity (e.g., different sample volume, adjustments in the incubation time, addition of cations or chelators to the reaction, different protocol

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adjustments in the software such as filters or gain). However, initial screening protocol used in Theralia R&D Laboratories is defined below.

- Add 8 µL sample (e.g., bacteria culture supernatant) into two different Eppendorf tubes.
- Add 1 µL PBS-/- into one tube and PBS+/+ into another and mix by pipetting.
- Add 1 µL working solution of the fluorescent oligonucleotide probe into each tube and mix by pipetting.
- Use PBS as sample in a separate tube for the background control of the probe.
- Prepare each reaction as triplicated.
- Incubate the mixture at 36±1°C for 1 hour.
- Add 295 µL stop solution (10 mM EDTA in PBS-/-) and mix by pipetting.
- Distribute this into three different wells of a 96-well black flat-bottom microplate as 95 µL
- Place the plate carefully into the plate reader (e.g., BioTek Synergy HTX multimode plate reader).
- Read the fluorescence intensity using the protocol defined in the software connected to the plate reader (e.g., Gen5) for initial screening (e.g., excitation wavelength: 485 nm / emission wavelength: 528 nm).
- Export the results to an excel file.

3.3. Evaluation of the results

- Calculate the average and SD values of the triplicates for each reaction.
- Calculate the fold differences for each probe by dividing the FI of the sample to the FI of the background (FD=sample FI / background FI)
- Prepare graphics for visualization of the results.
- Report the results as described in the **SOP-TL-DATA.**

4. Purpose/Objective

The objective of this SOP is to guide Theralia staff on performing Nuclease Activity Assay for bacterial nucleases in Theralia R&D Laboratories.

5. Definition / Explanation of key terms

- SOP – Standard Operating Procedure
- R&D – Research and Development
- CTO – Chief Technology Officer
- CSO – Chief Scientific Officer
- COO – Chief Operations Officer
- PPE – Personal Protective Equipment
- BSL-2 – Biosafety Level 2
- NAA – Nuclease Activity Assay
- EDTA – Ethylenediaminetetraacetic acid
- PBS-/- – Phosphate Buffered Saline

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- PBS+/+ – Phosphate Buffered Saline supplemented with Calcium and Magnesium
- CFU – Colony Forming Unit
- SD – Standard Deviation
- FI – Fluorescence Intensity
- FD – Fold Difference
- µL – Microliter
- mL – Milliliter
- nm – Nanometer
- pmol – Picomol
- mM - Millimolar

6. Responsibilities

This SOP was prepared by CTO, Baris Ata Borsa, revised by the CSO, Frank Hernandez and approved by the COO, Luiza I Hernandez

7. Equipment/Material

- BSL-2 biosafety cabinet
- Inoculation loops
- Bunsen burner
- 96-well black flat-bottom plates with lid
- Glass flasks
- Gloves
- Pipettes
- Pipette tips
- Incubator with shaker
- Plate reader

8. Caution/Warning

- Use BSL-2 biosafety cabinet for all the work that involves bacteria.
- Use PPE according to **SOP-TL-BSL2**.
- Clean the working area before and after the work.

9. References

- F.J. Hernandez, L. Huang, M.E. Olson, et. al. Noninvasive imaging of Staphylococcus aureus infections with a nuclease-activated probe, Nat. Med. 20 (2014) 301–306.

10. Version Control

Original version of the **SOP-TL-NAAB.v01**. Check **Table 1**.

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11.Attachments/Appendices

Appendix I:

- Bunsen burner manual
- Plate reader manual
- BSL-2 biosafety cabinet manual
- Incubator manual

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